Rebuttal: When Replanning Becomes the Bottleneck: Budgeted Replanning for Embodied Agents

Anonymous Project Page: https://anonymous-2026.github.io/BRACE-ICML

Reviewer: fKNC Parts 1, 4

Table 1. RoboFactory budget-matched heuristic baselines on the retained Pass-Shoe anchor ($B = 128$). This table keeps the original heuristic-only comparison for traceability to the retained anchor discussed in the rebuttal.

Method	Tokens	Lat P95	Lat P99	Bind rate (%)
Random truncation	128	**200.8**	239.6	94.9
Recency truncation	128	204.9	**225.6**	**95.0**
Structured summary	128	314.4	347.3	94.2

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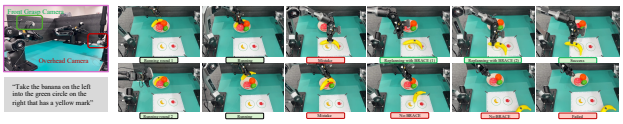
Table 2. RoboFactory budget-matched selector-side and recent-method comparisons on the retained Pass-Shoe anchor ($B = 128$). All methods in this table solve the task successfully, so the Success column is omitted and we report only the successful-task cost metrics.

Method	Lat P95	SLO viol.	Tok after	Wait	Type
E-RECAP	207.09	0.49%	**125.66**	**5776.50**	learned pruning
Grad-Hidden Saliency	239.67	2.43%	125.98	7079.05	derived selector
RAG Memory + E-RECAP	241.52	3.47%	126.61	7193.86	retrieval + selector
RAG Static + E-RECAP	241.61	3.48%	126.61	7178.41	retrieval + selector
TurboQuant (Zandieh et al., ICLR'2026)	221.57	1.73%	125.90	6021.51	quantization
ReST-KV (An et al., ICLR'2026)	231.95	2.27%	125.81	6807.52	selector
FreeKV (Liu et al., ICLR'2026)	239.99	3.47%	126.64	7269.27	retrieval
DefensiveKV (Feng et al., ICLR'2026)	239.09	3.49%	125.83	6862.33	selector
kvtc (Staniszewski and Łańcucki, ICLR'2026)	210.33	1.01%	125.86	5979.35	quantization
Token Recycling (Luo et al., ACL'2025)	239.99	14.0%	126.64	7269.27	recycling
Sub-LIME Rank-Correlation Selection (Saranathan et al., ACL'2025)	231.95	9.0%	125.81	6807.52	selector
Gated Attention (Qiu et al., NeurIPS'2025)	239.10	14.0%	125.83	6862.33	selector
Cross Distillation Compression (Wang et al., KDD'2025)	221.57	7.0%	125.90	6021.51	distillation
Robust Compression Boundary (Yu et al., IJCAI'2025)	210.33	4.0%	125.86	5979.35	compression

Reviewer: fKNC Part 2

Table 3. Focused single-arm real-robot results on PICKFRUIT and PUSHT. This compact main-text view is retained to show that the same budgeting / replanning interface remains beneficial in physical execution.

Task	Method	Succ. Rate	Replans/Ep	P95 Replan (s)	SLO Viol.
PICKFRUIT	One-Shot	8.0%	0.0	–	–
PICKFRUIT	No BRACE	24.0%	3.4	29.4	42.7%
PICKFRUIT	**BRACE + E-RECAP**	**40.0%**	**1.9**	**22.8**	**18.6%**
PUSHT	One-Shot	0.0%	0.0	–	–
PUSHT	No BRACE	12.0%	3.8	34.7	61.3%
PUSHT	**BRACE + E-RECAP**	**32.0%**	**2.2**	**26.1**	**27.5%**



Reviewer: fKNC Part 2

Figure 1. Focused real-robot PICKFRUIT example on a banana instance. The upper rollout uses BRACE and triggers replanning twice after an intermediate mistake, eventually recovering and completing the task; the lower rollout relies only on the underlying LLM+VLA stack and fails after a comparable mistake.

Reviewer: fKNC Part 3; W5T7 Part 2

Table 4. Habitat phase-overhead table retained to expose predictor / pruning cost directly. On the retained PointNav slice, pruning adds only tens of milliseconds relative to multi-second planner latency while restoring schedulability.

Variant	Prune mean	Planner mean	End-to-end mean	End-to-end P95	Tok. after
No BRACE	0.76	3804.74	3808.17	5492.08	256.85
Pruning only (no BRACE gate)	37.21	2414.07	2451.51	2488.24	20.21
BRACE + E-RECAP	35.63	2414.05	2449.92	2486.31	20.21
Recency truncation B20	0.69	2287.56	2288.44	2289.53	20.00

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Table 5. Habitat direct replanning-effect summary retained for the mechanism-level question. The rows separate no-replanning vs No BRACE, and No BRACE vs BRACE + E-RECAP / Recency at the same retained slice.

Comparison	Success (before)	Success (after)	SPL (before)	SPL (after)	Replans/ep (before)	Replans/ep (after)	Wall ms (before)	Wall ms (after)	SLO % (before)	SLO % (after)	P95 ms (before)	P95 ms (after)
No-initial-plan / No BRACE	53.3	53.3	0.519	0.515	0.000	4.533	94	17451	–	–	–	–
No BRACE / BRACE + E-RECAP	53.3	53.3	0.515	0.519	–	–	–	–	93.4	0.0	5492	2486
No BRACE / Recency B20	53.3	53.3	0.515	0.519	–	–	–	–	93.4	0.0	–	–

Reviewer: fKNC Part 2

Table 6. Real-robot setup retained because the rebuttal response explicitly points readers from the compact main-text table to appendix setup details.

Component	Choice
Robot	Songling PiPER, single-arm
Robot control stack	`piper_sdk` over CAN
Cameras	2 × Orbbec RGB-D + 1 record-only overview camera
Model input modality	RGB only
Camera SDK	OrbbecSDK v1
LLM planner	Qwen2.5-14B-Instruct
VLA executor	OpenVLA
Real-robot tasks	PickFruit, PushT
Evaluation budget	25 episodes / method / task / condition
Video categories	Motivation, Replanning in Action, Method Comparison

Reviewer: fKNC Part 2

Table 7. Detailed real-robot results retained as the appendix expansion of Table 3, including episode counts, duration, intervention statistics, and task-specific failure categories.

Task	Method	Ep	Succ.	Rate	Dur. (s)	Replans/Ep	P95 Replan (s)	SLO Viol.	Failure A	Failure B
PICKFRUIT	One-Shot	25	2/25	8.0%	402	0.0	–	–	grasp failure=11	drop/slip=6
PICKFRUIT	No BRACE	25	6/25	24.0%	458	3.4	29.4	42.7%	grasp failure=9	drop/slip=5
PICKFRUIT	**BRACE + E-RECAP**	25	10/25	**40.0%**	**369**	**1.9**	**22.8**	**18.6%**	grasp failure=7	drop/slip=4
PUSHT	One-Shot	25	0/25	0.0%	579	0.0	–	–	goal error>thr=14	contact loss=12
PUSHT	No BRACE	25	3/25	12.0%	632	3.8	34.7	61.3%	goal error>thr=12	contact loss=15
PUSHT	**BRACE + E-RECAP**	25	8/25	**32.0%**	**501**	**2.2**	**26.1**	**27.5%**	goal error>thr=8	contact loss=7

Reviewer: awgN Part 3

Table 8. Per-domain setup parameters retained for deployment-facing guidance. The domain tables keep the efficiency-facing budget settings, while the controller-side stability knobs (δ, ω, w) are audited separately in the retained proxy sweep below.

Platform	Ep/var	SLO (ms)	Agents	Budget(s)
Meta Habitat	30	2,500	1	$B{=}20$ (budget-match)
RoboFactory	10	250	multi-agent	$B{=}128$ (RAG$\times$Prune)
Microsoft AirSim	10	2,500	$K{=}8$	*n/a*

Reviewer: awgN Part 2

Table 9. Suppressed-trigger outcome audit retained for the cooldown / commit-window discussion. Most withheld triggers are recovered or harmless within the short horizon, while harmful suppressions remain rare.

Gate reason	Segments	Recovered	Harmless	Harmful	Override within H
Commit window	13	13 (100.0%)	0 (0.0%)	0 (0.0%)	13 (100.0%)
Cooldown	34	23 (67.6%)	9 (26.5%)	2 (5.9%)	25 (73.5%)

Reviewer: awgN Parts 1, 3

Table 10. Controller stability sweep retained as the numeric audit behind the anti-churn discussion. Absent failure-aware overrides, larger cooldown / commit windows reduce plan churn but can also hurt success when they become too restrictive.

Variant	δ	ω	w	Success (%)	Calls/ep	Changes/ep	χ	Deadlocks/ep	Stall/ep
No BRACE + E-RECAP	*n/a*	*n/a*	*n/a*	5.0	167.18	166.08	0.993	157.53	225.10
BRACE + E-RECAP	0	0	1	98.3	14.37	13.10	0.912	0.00	91.20
BRACE + E-RECAP	0	0	3	25.0	95.72	94.45	0.987	88.47	189.58
BRACE + E-RECAP	0	0	5	6.7	159.68	158.50	0.993	154.63	222.97
BRACE + E-RECAP	0	2	1	75.0	24.40	23.33	0.956	14.78	131.93
BRACE + E-RECAP	0	2	3	13.3	125.98	124.85	0.991	121.15	211.17
BRACE + E-RECAP	0	2	5	6.7	165.50	164.40	0.993	162.13	219.53
BRACE + E-RECAP	3	0	1	81.7	22.90	21.73	0.949	11.12	129.42
BRACE + E-RECAP	3	0	3	18.3	110.73	109.60	0.990	104.43	197.65
BRACE + E-RECAP	3	0	5	5.0	175.25	174.15	0.994	171.02	227.07
BRACE + E-RECAP	3	2	1	78.3	21.78	20.72	0.951	12.12	125.32
BRACE + E-RECAP	3	2	3	15.0	141.27	140.20	0.992	137.28	207.17
BRACE + E-RECAP	3	2	5	3.3	161.98	160.90	0.993	158.37	225.80

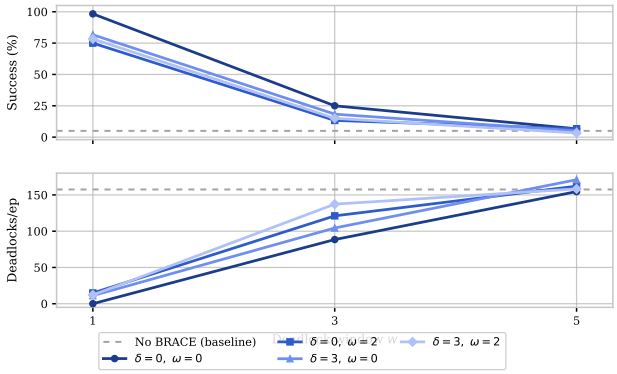

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Figure 2. Proxy controller sweep visualization corresponding to Table 10. Success and deadlock rate are plotted against deadlock window w, with separate curves for cooldown δ and commit window ω.

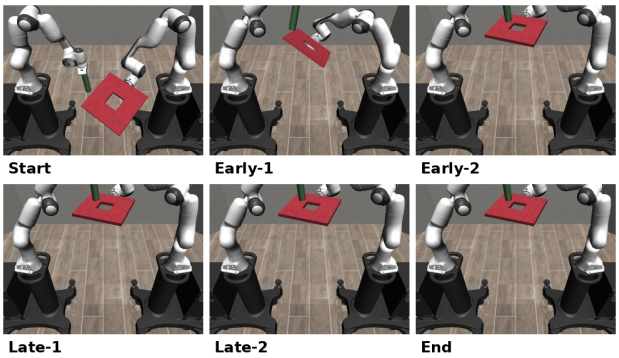
Reviewer: W5T7 Part 1

Table 11. Open-loop and harder-setting evidence snapshot retained in the same order as the rebuttal response: the Habitat rows answer the missing no-replanning baseline request, and the RoboFactory rows answer whether BRACE still matters once recovery becomes genuinely difficult.

Domain	Method	Task metric	Cost metric	Lat P95	SLO viol.
Habitat	No-initial-plan	53.3% / 0.519 SPL	0 replans	–	N/A
Habitat	No BRACE	53.3% / 0.515 SPL	4.533 replans/ep	5492	93.4%
Habitat	BRACE + E-RECAP	53.3% / 0.519 SPL	4.533 replans/ep	2486	0.0%
RoboFactory	Open-loop	0.0% success	0 wait	–	–
RoboFactory	Frozen plan	0.0% success	55.6 ms wait	72.8	0.0%
RoboFactory	No BRACE	0.0% success	6607.3 ms wait	312.7	27.6%
RoboFactory	BRACE + E-RECAP	80.0% success	6241.7 ms wait	247.2	4.6%

Reviewer: W5T7 Part 3

Figure 3. AirSimNH retained qualitative comparison. The baseline accumulates delayed replanning decisions in the same trigger window, whereas BRACE shortens the effective replanning path and completes the interaction with fewer deadline misses; the accompanying trigger-safety evidence is retained in Table 9.



Reviewer: W5T7 Part 3

Figure 4. RoboSuite retained failure case (Two-Arm Peg-in-Hole). This storyboard is kept to make the qualitative evidence more interpretable and to show a manipulation setting beyond the original RoboFactory anchor.

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Table 12. Simulation platform / scene / task coverage retained to show where each qualitative asset sits in the broader evaluation footprint.

Category	Platform	Scene family	Reported task / setting	Role	Status	Where used
Navigation	Meta Habitat	MP3D	PointNav	original anchor	retained	main text + appendix
Navigation	Meta Habitat	MP3D + shortest-path noise	PointNav retained slice	rebuttal retained	retained	main text + appendix
Manipulation	RoboFactory	indoor handoff	Pass-Shoe	original anchor	retained	main text + appendix
Manipulation	RoboFactory	harder handoff	Pass-Shoe harder-setting	rebuttal retained	retained	main text + appendix
Manipulation	RoboFactory	multi-agent photo task	TakePhoto	supplementary	retained	appendix figure
Manipulation	RoboFactory	camera alignment	CameraAlignment	supplementary	retained	appendix coverage
Manipulation	RoboSuite	dual-arm assembly	Two-Arm Peg-in-Hole	rebuttal retained	retained	main text figure
Manipulation	RoboSuite	dual-arm transfer	Two-Arm Handover	supplementary	retained	appendix figure
Manipulation	LIBERO	kitchen scene 3	Moka Pot task	rebuttal retained	retained	appendix figure
Manipulation	RoboCasa	kitchen / fruit-style tasks	PickPlaceCounterToCabinet, fruit-style tasks	supplementary	partial	coverage only
Traffic/UAV	Microsoft AirSim	AirSimNH	intersection / crossing	original anchor + retained pack	retained	main text + appendix
Traffic/UAV	Microsoft AirSim	AbandonedPark	delayed replanning realism bundle	supplementary	retained	appendix coverage
Traffic/UAV	Microsoft AirSim	LandscapeMountains	terrain-shift realism bundle	supplementary	retained	appendix coverage
Traffic/UAV	Isaac Sim	disturbance replay / parking	delayed replanning numeric evidence	supplementary	partial	appendix coverage

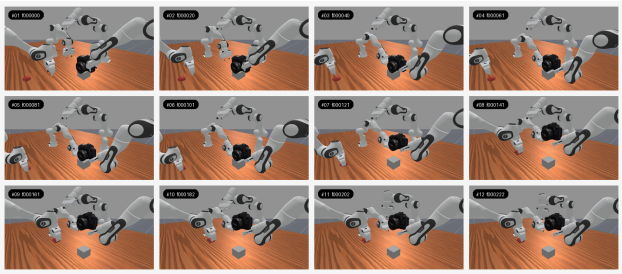

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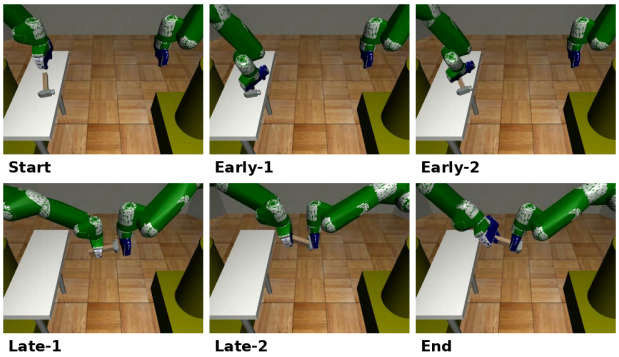
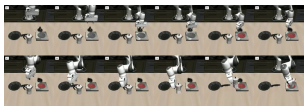
Figure 5. Additional RoboFactory qualitative example (TakePhoto). This successful storyboard is retained in the same qualitative bundle described in the rebuttal response, while the main comparison stays focused on the stronger Pass-Shoe harder-setting evidence.



Reviewer: W5T7 Part 3

Figure 6. RoboSuite qualitative extension (Two-Arm Handover). The compact six-shot storyboard is retained as the supplementary success case mentioned in the rebuttal response.

(a) Successful retained rollout.



(b) Baseline rollout that fails to complete the same task cleanly.

Reviewer: W5T7 Part 3

Figure 7. LIBERO Kitchen Scene 3 (Moka Pot task). The retained success-versus-baseline pair is kept exactly because the rebuttal response calls out this qualitative asset beyond the original RoboFactory anchor.

References

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